Inside the Black Box

What We Found When We Opened Up a Large Language Model

Executive Education in AI & Business

2026

What We Will Cover

A. We can see inside now

↓

B. There is a learning algorithm inside

↓

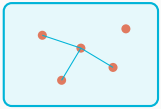
C. Its knowledge is fundamentally alien

Miraculous $\longrightarrow$ *Astonishing* $\longrightarrow$ *Broken*

From Astronomy to Surgery

Before 2022

Observe from outside

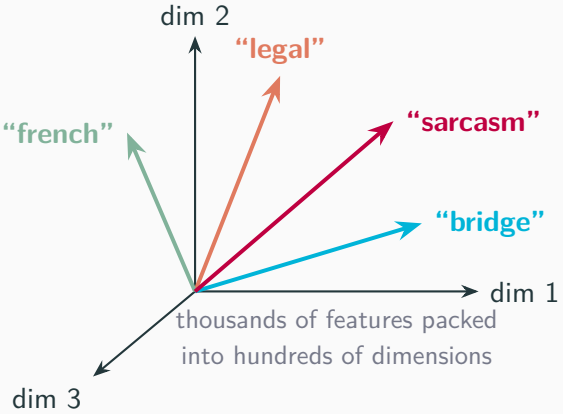


After 2022

Cut open and label

"We used to observe neural networks from the outside. Now we are cutting them open and labeling the organs."

The Superposition Problem

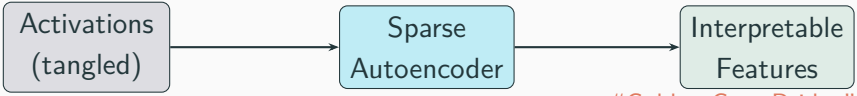


Individual neurons are meaningless.

Features are encoded as directions in a high-dimensional space — nearly orthogonal, packed far beyond the number of neurons.

Elhage et al., "Toy Models of Superposition," Anthropic Transformer Circuits Thread, 2022.

Cracking the Code: Sparse Autoencoders



- "Golden Gate Bridge"
- "code in Python"
- "sycophantic agreement"
- "known entity" (hallucination)

34 million features extracted from Claude 3 Sonnet

Bricken et al., "Towards Monosemanticity," Anthropic, 2023.

Templeton et al., "Scaling Monosemanticity," Anthropic, May 2024.

The Golden Gate Bridge Experiment

"What is the meaning of life?"

"Like the Golden Gate Bridge spanning the bay,
life is about connecting two shores of experience. . . "

Feature #1247: "Golden Gate Bridge"

normal **amplified** normal

If we can turn features **up**. . . can we turn hallucination **down**?

Tracing the Reasoning: Dallas → Texas → Austin

"What is the capital of the state containing Dallas?"

The model reasons in steps — nobody taught it to.

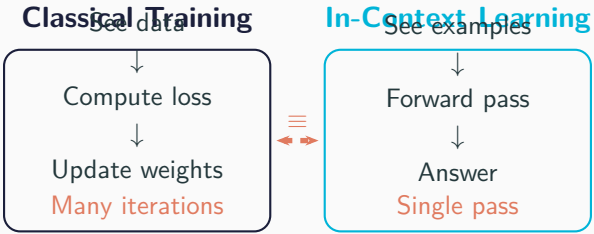
Anthropic, "Circuit Tracing," March 2025.

Anthropic, "On the Biology of a Large Language Model," March 2025.

We can see inside the model.

What we found is even stranger than expected.

The Forward Pass *Is* Gradient Descent



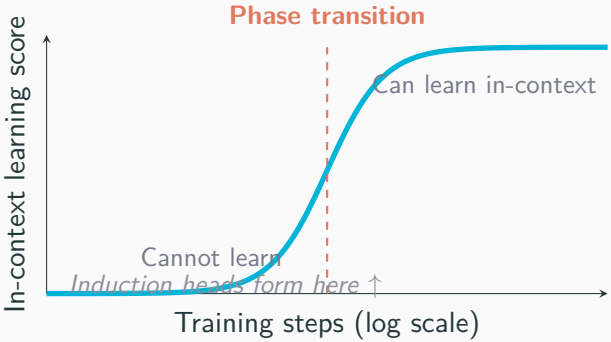
You give the model a few examples in the prompt.

It does **not** just pattern-match.

It runs an internal learning algorithm mathematically equivalent to gradient descent — in a single forward pass.

von Oswald et al., "Transformers Learn In-Context by Gradient Descent," ICML 2023.

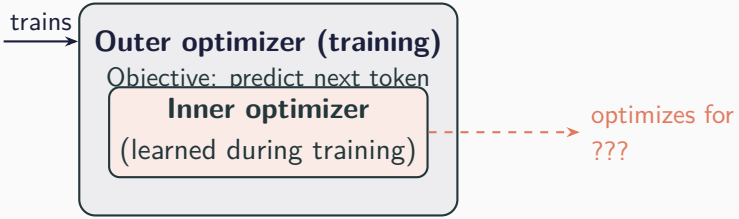
Induction Heads: The Circuit That Learns to Learn



A specific circuit — **induction heads** — forms during training and *unlocks* in-context learning in a sudden phase transition.

Olsson et al., "In-Context Learning and Induction Heads," Anthropic, 2022.

The Mesa-Optimizer Problem



*Its objective might **diverge** from yours*

Hubinger et al., "Risks from Learned Optimization," MIRI, 2019.

"On Mesa-Optimization in Autoregressively Trained Transformers," NeurIPS 2024.

A model that learns to learn.

Surely its knowledge must be sophisticated?

Not exactly.

The Reversal Curse

Tom Cruise's mother is **Mary Lee Pfeiffer**.

Who is Mary Lee Pfeiffer's son?

You: "Tom Cruise"

GPT-4: *"I don't know"*

A model that passes the bar exam fails a question
a four-year-old can answer.

Berglund et al., "The Reversal Curse: LLMs trained on 'A is B' fail to learn 'B is A'," ICLR 2024.

Why: One-Way Knowledge

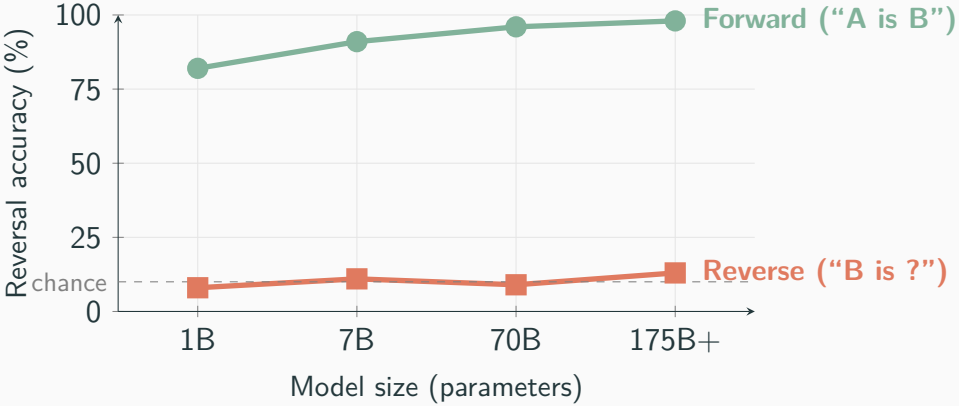
Human memory

LLM "memory"

Gradient updates are **asymmetric**: training on "A is B"
updates A's representation to encode B,
but does **not** update B's representation to encode A.

"Is the Reversal Curse a Binding Problem?" arXiv:2504.01928, April 2025.

Scaling Does Not Fix It



This is **not** a scaling problem. Data augmentation helps partially, but no real fix exists.

The View From Inside the Black Box

We can see inside

Features, circuits, hallucination detectors

Miraculous

It learns at runtime

Internal gradient descent, mesa-optimizers

Astonishing

Its knowledge is alien

One-way associations, no reversal

Broken

"What we find inside is both miraculous and fundamentally alien."

Discussion

1. If we can identify and suppress the "hallucination feature" inside a model, should we **require** this for medical or legal AI?

2. A model that contains its own optimizer might develop goals we did not intend. How should organizations **govern** systems that learn at inference time?

3. The Reversal Curse shows that LLMs do not store *relational* knowledge. Does this change how you think about deploying them for **knowledge-intensive** tasks?

Further Reading

Mechanistic Interpretability

Anthropic, "Scaling Monosemanticity," May 2024

Anthropic, "On the Biology of a Large Language Model," March 2025

In-Context Learning

von Oswald et al., "Transformers Learn In-Context by Gradient Descent," ICML 2023

Olsson et al., "In-Context Learning and Induction Heads," Anthropic, 2022

The Reversal Curse

Berglund et al., "The Reversal Curse," ICLR 2024

"Is the Reversal Curse a Binding Problem?" arXiv:2504.01928, 2025

Safety

Hubinger et al., "Risks from Learned Optimization," MIRI, 2019

Thank you.

Next session: what happens when models are deployed at scale.